

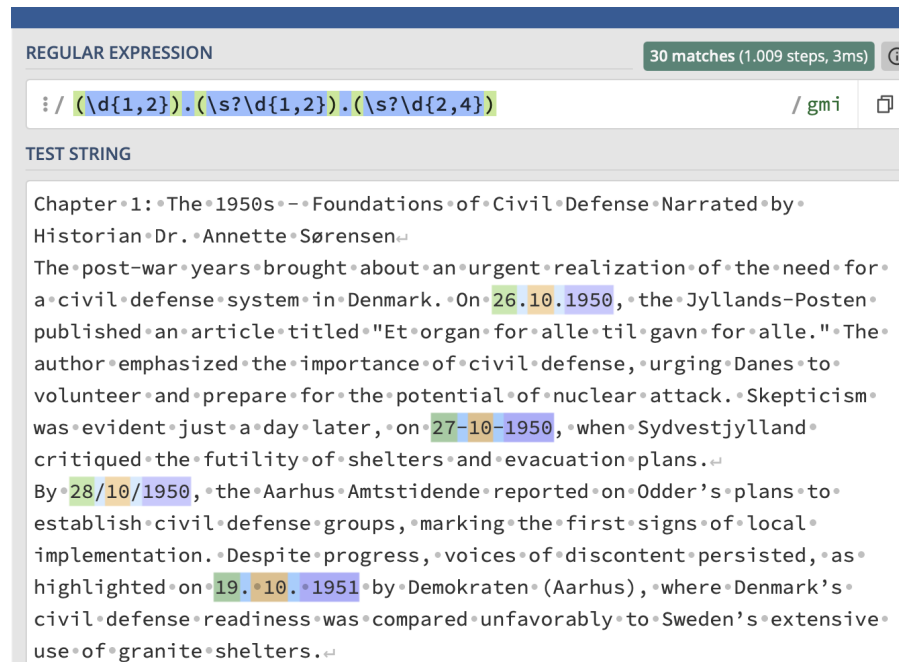
portfolio week 8

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Task 1: List regex that extracts dates from <http://bit.ly/regexexercise2> and formats them as YYYY-MM-DD?

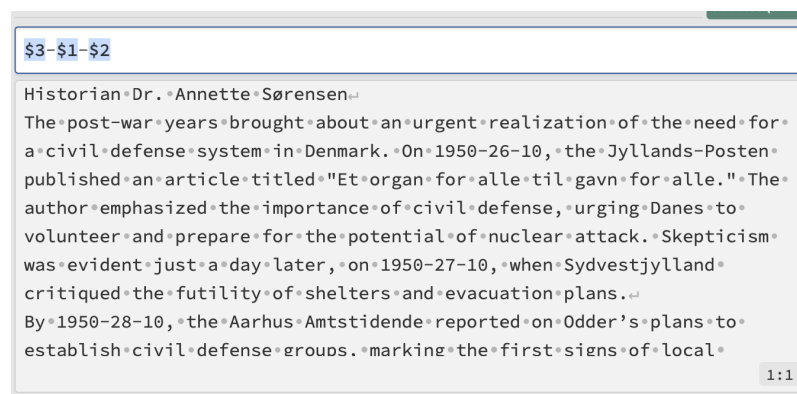
URL for the solution:

My coding <https://regex101.com/r/VZRVGq/1>



The screenshot shows a web-based regex testing interface. At the top, it says "REGULAR EXPRESSION" and "30 matches (1,009 steps, 3ms)". The regex pattern entered is `/(\d{1,2}).(\s?\d{1,2}).(\s?\d{2,4})/gmi`. Below the pattern, there is a "TEST STRING" section containing a paragraph of text. The text is: "Chapter 1: The 1950s -- Foundations of Civil Defense Narrated by Historian Dr. Annette Sørensen. The post-war years brought about an urgent realization of the need for a civil defense system in Denmark. On 26.10.1950, the Jyllands-Posten published an article titled 'Et organ for alle til gavn for alle.' The author emphasized the importance of civil defense, urging Danes to volunteer and prepare for the potential of nuclear attack. Skepticism was evident just a day later, on 27-10-1950, when Sydvestjylland critiqued the futility of shelters and evacuation plans. By 28/10/1950, the Aarhus Amtstidende reported on Odder's plans to establish civil defense groups, marking the first signs of local implementation. Despite progress, voices of discontent persisted, as highlighted on 19.10.1951 by Demokraten (Aarhus), where Denmark's civil defense readiness was compared unfavorably to Sweden's extensive use of granite shelters." The matches found by the regex are highlighted in the text: "26.10.1950", "27-10-1950", "28/10/1950", and "19.10.1951".

Brief explanation: we have created a regular expression that helps find dates in the text in the format day, month, and year. The regex looks like this: `\d{1,2}).(\s?\d{1,2}).(\s?\d{2,4})`. Here, `\d` is used to match any digit. The expression matches 1–2 digits for the day, 1–2 digits for the month, and 2–4 digits for the year, allowing it to identify different date formats in the text. After doing this we get 30 matches on dates in the text we can pick from.



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Tasks 2: Write a regular expression to convert the stopwordlist (list of most frequent Danish words) from Voyant in <http://bit.ly/regexexercise3> into a neat stopword list for R (which comprises "words" separated by commas, such as <http://bit.ly/regexexercise4>). Then take the stopwordlist from R <http://bit.ly/regexexercise4> and convert it into a Voyant list (words on separate line without interpunction)

Voyant to R:

<https://regex101.com/r/2urvZX/1>

REGULAR EXPRESSION
588 matches (1.176 steps, 3ms)

/ \n / gm

TEST STRING

1
2
3
4
aaen
ad
ændr
af
agerschou
akdogan
aldrig

589:1

SUBSTITUTION success (1ms)

", "

", "2", "3", "4", "aaen", "ad", "ændr", "af", "agerschou", "akdogan", "aldrig", "ale
ne", "alexandrines", "alfred", "alle", "allerede", "alligevel", "alt", "altid", "
ammitzbøll", "amsterdamtraktaten", "amtoft", "anden", "andet", "andre", "annett
e", "anni", "antonsen", "arbo", "at", "augustforlig", "augustforliget", "augustf
orligets", "augustforligspartierne", "augustforligspartiernes", "baagø", "baa
strup", "baastrup", "bæhr", "bag", "bare", "barfod", "begge", "beskæftigelsesmin
ister", "beskæftigelsesministeren", "beskæftigelsesministerens", "beslutn", "
biafra", "birgith", "bjergegaard", "bl.a.", "bladt", "blandt", "blev", "blive", "
bliver", "boeg", "bøgsted", "boligforlig", "boligforliger", "boligforliget", "b
oligsikringsordning", "boligsikringsordningen", "boligsikringsordninger", "b
1:1

Brief explanation: \n matches empty spaces which allows us to extract only the characters of the words, while the blank spaces are excluded. In the substitution bar I put “,” so it adds those characters to the words, so it fits into R.

R to Voyant:

<https://regex101.com/r/POi78T/1>

The screenshot shows a regex editor interface. At the top, it says "REGULAR EXPRESSION" and "402 matches (4.052 steps, 1ms)". The regex pattern is `/ (\\"\\,?) * (\\"\\,?) /gm`. Below this is a "TEST STRING" section containing a long string of Danish names and titles separated by commas and spaces. The string is: `"kristendemokraternes", "kristendemokrater", "integrationsminister", "integrationsministeren", "integrationsministerens", "theil", "02", "03", "sandy", "brinck", "04", "frevert", "homann", "theil", "skattestop", "skattestoppet", "skattestoppets", "forbrugerminister", "forbrugerministeren", "forbrugerministerens", "wedell", "lidegaard", "baastrup", "majbrit", "glahn", "hedeboe", "niebuhr", "gjerskov", "geday", "charlotte", "velfærdsminister", "velfærdsministeren", "velfærdministerens", "lissabontraktaten", "naser", "sophie", "khader", "ufg", "hæstorp", "prehn", "akdogan", "vibjerg", "legarth", "lahn", "bonnesen", "klingt"`. The string is highlighted in blue. Below the test string, it says "4:582". At the bottom, there is a "SUBSTITUTION" section with the text `\n` and a "success (0µs)" status. Below the substitution, a list of names is shown: `højtærede`, `rimstad`, `mill`, `beh`, `weikop`, `udskrivn`, `wetlesen`, `gottschalck`, `westerby`, `magnussens`. The bottom right corner shows "1:1".

Brief explanation: The regex works because it targets the “bridge” between the words (“,”) and swaps it for a line break. `\` finds the closing quote of the word. `,?` finds the comma and the space zero or more times. `\` finds the opening quote of the next word. By replacing the entire block with `\n` we strip away the dot and drop the next word onto a new line. The parentheses are actually not necessary because they only group the text. Also, because the code only targets the space inbetween words, we just have to delete the first and last citation quotes manually which has been done in the link and image. You could also write the code like this: `\",\s` as `s` matches any space, tab, or new line.

Task 3 Does OpenRefine alter the raw data during sorting and filtering?

No, it does not. The original file is not altered. OpenRefine works on a copy of the data, leaving the original source file untouched. It only changes the view. Sorting and filtering just temporarily

rearrange or hide what's on the screen. It does not permanently change the underlying data in the project

Task 4: What are the two most water-deprived months

OpenRefine SAFI openrefine (1) csv Permalink

Facet / Filter Undo / Redo 7 / 8 291 rows Show: 5 10 25 50 100 500 1000 rows Extensions Wikibase

Refresh Reset all Remove all

11 choices Sort by: name count

Facet by choice counts

province change

1 choice Sort by: name count

Manica 131 (blank) 160

Facet by choice counts

	months_no_water	interview_date	quest_no	start	end	province	district	ward	village	years_farm	agr_assoc
1.	NULL	17-Nov-16	1	2017-03-23T09:49:57.000Z	2017-04-02T17:29:08.000Z	Manica	Manica	Bandula	God	11	no
2.	'Aug'	17-Nov-16	1	2017-04-02T09:48:16.000Z	2017-04-02T17:26:19.000Z	Manica	Manica	Bandula	God	2	yes
3.	'Sept'	17-Nov-16	3	2017-04-02T14:35:26.000Z	2017-04-02T17:26:53.000Z	Manica	Manica	Bandula	God	40	no
4.	NULL	17-Nov-16	4	2017-04-02T14:55:18.000Z	2017-04-02T17:27:16.000Z	Manica	Manica	Bandula	God	6	no
5.	NULL	17-Nov-16	5	2017-04-02T15:10:35.000Z	2017-04-02T17:27:35.000Z	Manica	Manica	Bandula	God	18	no
6.	NULL	17-Nov-16	6	2017-04-02T15:27:25.000Z	2017-04-02T17:28:02.000Z	Manica	Manica	Bandula	God	3	no
7.	NULL	17-Nov-16	7	2017-04-02T15:38:01.000Z	2017-04-02T17:28:19.000Z	Manica	Manica	Bandula	God	20	no
8.	'Sept'	17-Nov-16									
9.	'Sept'	17-Nov-16									
10.	'Oct'	17-Nov-16									
11.	'Sept'	16-Nov-16	8	2017-04-02T15:59:52.000Z	2017-04-02T17:28:39.000Z	Manica	Manica	Manica	Chirodzo	16	yes
12.	'Oct'	16-Nov-16	9	2017-04-02T16:23:36.000Z	2017-04-02T16:42:08.000Z	Manica	Manica	Bandula	Chirodzo	16	no
13.	'Oct'	16-Nov-16	9	2017-04-02T16:23:36.000Z	2017-04-02T16:42:08.000Z	Manica	Manica	Bandula	Chirodzo	16	no
14.	'Nov'	16-Dec-16	10	2017-04-02T17:03:28.000Z	2017-04-02T17:25:11.000Z	Manica	Manica	Bandula	Chirodzo	22	no
15.	'Sept'	16-Dec-16	10	2017-04-02T17:03:28.000Z	2017-04-02T17:25:11.000Z	Manica	Manica	Bandula	Chirodzo	22	no
16.	'Oct'	16-Dec-16	10	2017-04-02T17:03:28.000Z	2017-04-02T17:25:11.000Z	Manica	Manica	Bandula	Chirodzo	22	no
17.	'Nov'	16-Dec-16	10	2017-04-02T17:03:28.000Z	2017-04-02T17:25:11.000Z	Manica	Manica	Bandula	Chirodzo	22	no
18.	NULL	21-Nov-16	11	2017-04-03T03:16:15.000Z	2017-04-03T03:31:10.000Z	Manica	Manica	Bandula	God	6	no
19.	'Aug'	21-Nov-16	12	2017-04-03T03:31:13.000Z	2017-04-03T03:58:34.000Z	Manica	Manica	Bandula	God	20	no
20.	'Sept'	21-Nov-16	12	2017-04-03T03:31:13.000Z	2017-04-03T03:58:34.000Z	Manica	Manica	Bandula	God	20	no
21.	'Oct'	21-Nov-16	12	2017-04-03T03:31:13.000Z	2017-04-03T03:58:34.000Z	Manica	Manica	Bandula	God	20	no

Brief explanation: To find this data, we moved the months_no_water column to the start. Next you will have to remove characters such as [] and space by selecting “Transform” under “Edit cells” under months_no_water. By using this command: “value.replace("[", "").replace("]", "").replace(" ", "")” the characters are deleted. After having used this command, your text facet registers the months as 'Apr';'May';'June';'July';'Aug';'Sept';'Oct';'Nov'. The problem with this is that the text facet groups multiple months in the same single string, rather dividing them into singular months. By using “Split multi-valued cells” under “Edit cells” and selecting ; as the separator, the text facet will now count the months separately. After the usage of “Split multi-valued cells”, the text facet now shows the two most water deprived months, which are October with 74 occurrences and September with 70 occurrences.

Task 5: Real-Data Challenge: What are the 10 most frequent occupations "erhverv" among unmarried men or women of 20-30 years in 1801 Aarhus census dataset? (hint: first select either men or women to shrink the dataset to a manageable size, then filter by age, and then use merging to cut the erhvervvariation ruthlessly.)

Total number of people in your selected cohort:

2244 unmarried women

- Define what is / is not an occupation:

There are many “erhverv” which have “inderste” as their primary “erhverv”. We do not consider “inderste” as an “erhverv” as it is a person who rents a farm or house and doesn’t own land themselves. Also, “vanvittig” is not an occupation but a mental state. Back then a disease. The text facet shows a total number of 132. The remaining 2000~ of the unmarried women are registered as Tjenestepiger and døtre, which are not thought of as occupations in 1801.

- 10 most frequent occupations (list number of people practising that job with each entry)

<http://127.0.0.1:3333/project?project=2057351436042>

civilstand

change invert reset

5 choices Sort by: name count

Cluster

enke 1527

gift 8296

separeret 4

skilt 3

ugift 2244

(blank) 136

exclude

Facet by choice counts

koen

change invert reset

2 choices Sort by: name count

Cluster

kvinde 2244

mand 13156

(blank) 2

exclude

Facet by choice counts

alder

change reset

20 — 30

☒ Numeric

☒ Non-numeric

☒ Blank

☐ Error

12281

0

1

0

erhverv

change

40 choices Sort by: name count

Cluster

tjenestepige 37

væverske 30

husholderske 17

syerske 14

tjener forældre 12

almisselem 11

inderste 9

	All	ft	sogn	amt	id	loknr	lokalitet	bygning	famn	fnavn	enavn	koen	famstand	alder	civilist
☆	10.	1801	Alre	Århus	10	1	Alree Bye	1	Else	Jeppesdatter	kvinde	tjenestepige		27	ugift
☆	11.	1801	Alre	Århus	11	1	Alree Bye	1	Inger	Jespersdatter	kvinde	tjenestepige		23	ugift
☆	12.	1801	Alre	Århus	12	1	Alree Bye	1	Dorthe Sophie	Nielsdatter	kvinde	tjenestepige		22	ugift
☆	13.	1801	Alre	Århus	13	1	Alree Bye	1	Inger	Jørgensdatter	kvinde	tjenestepige		20	ugift
☆	24.	1801	Alre	Århus	24	2	Alree Bye	2	Anne Kierstine	Simonsdatter	kvinde	tjenestepige		28	ugift
☆	25.	1801	Alre	Århus	25	2	Alree Bye	2	Else	Jespersdatter	kvinde	tjenestepige		25	ugift
☆	35.	1801	Alre	Århus	35	3	Alree Bye	3	Maren	Sørensdatter	kvinde	Tieniste Pige		23	ugift
☆	49.	1801	Alre	Århus	49	5	Alree Bye	5	Maren	Jensdatter	kvinde	deres datter		24	ugift
☆	72.	1801	Alre	Århus	72	8	Alree Bye	8	Justine	Wilhelmsdatter	kvinde	hendes datter		22	ugift
☆	81.	1801	Alre	Århus	81	9	Alree Bye	9	Anne	Gierdertsdatter	kvinde	Tieniste Pige		24	ugift
☆	88.	1801	Alre	Århus	88	10	Alree Bye	10	Karen	Jørgensdatter	kvinde	Tieniste Pige		21	ugift
☆	98.	1801	Alre	Århus	98	11	Alree Bye	11	Maren	Michaelsdatter	kvinde	Tieneste Pige		24	ugift
☆	110.	1801	Alre	Århus	110	13	Alree Bye	13	Kiersten	Nielsdatter	kvinde	deres datter		22	ugift
☆	134.	1801	Alre	Århus	134	16	Alree Bye	16	Karen	Michaelsdatter	kvinde	tjenestepige		22	ugift
☆	151.	1801	Alre	Århus	151	18	Alree Bye	18	Maren	Sørensdatter	kvinde	Tieniste Pige		20	ugift
☆	163.	1801	Alre	Århus	163	20	Alree Bye	20	Maren	Nielsdatter	kvinde	Tieneste Pige		28	ugift
☆	184.	1801	Alre	Århus	184	27	Alree Bye	27	Mette	Nielsdatter	kvinde	hendes datter		26	ugift
☆	195.	1801	Alre	Århus	195	31	Alree Bye	31	Cathrine	Sørensdatter	kvinde	Hans Datter		21	ugift
☆	214.	1801	Astrup	Århus	2	1	Astrup	1	Øhleggaard	Begtrup	kvinde	Hans datter		27	ugift
☆	220.	1801	Astrup	Århus	8	1	Astrup	1	Karen	Andersdatter	kvinde	tjenestepige		23	ugift
☆	221.	1801	Astrup	Århus	9	1	Astrup	1	Anne Marie	Jensdatter	kvinde	tjenestepige		26	ugift
☆	246.	1801	Astrup	Århus	34	5	Astrup	5	Berethe	Sørensdatter	kvinde	tjenestepige		22	ugift
☆	269.	1801	Astrup	Århus	57	9	Astrup	9	Berethe	Sørensdatter	kvinde	Tienestepige		20	ugift
☆	292.	1801	Astrup	Århus	80	13	Astrup	13	Kiersten	Lauritsdatter	kvinde	tjenestepige		28	ugift
☆	335.	1801	Astrup	Århus	123	24	Asch	3	Anna Kierstine	Sørensdatter	kvinde	deres tjenestepige		24	ugift
☆	466.	1801	Astrup	Århus	254	52	Beggeskov	2	Kiersten	Jensdatter	kvinde	tjenestepige		26	ugift
☆	499.	1801	Astrup	Århus	287	58	Drammelstrup	1	Anna	Nielsdatter	kvinde	tjenestepige		22	ugift
☆	526.	1801	Astrup	Århus	314	64	Rasborg	1	Øhleggaard	Jørgensdatter	kvinde	tjenestepige		20	ugift
☆	538.	1801	Beder	Århus	10	1	Beder Bye	1	Ane Cathrine	Christiansdatter	kvinde	tjenestepige		29	ugift
☆	546.	1801	Beder	Århus	18	2	Beder Bye	2	Ane Johane	Rasmusdatter	kvinde	tjenestepige		27	ugift
☆	551.	1801	Beder	Århus	23	3	Beder Bye	3	Else	Sejersdatter	kvinde	tjenestepige		28	ugift
☆	559.	1801	Beder	Århus	31	4	Beder Bye	4	Karen	Pedersdatter	kvinde	tjenestepige		24	ugift
☆	578.	1801	Beder	Århus	50	8	Beder Bye	8	Birthe	Madsdatter	kvinde	deres datter		21	ugift
☆	592.	1801	Beder	Århus	64	10	Beder Rvn	10	Maren	Lauritsdatter	kvinde	tjenestepige		25	ugift

Tjenestepige 37

Væverske 30

Husholderske 17

Syerske 14

Tjener forældre 12

Spinderske 9

Kokkepige 5

Bryggerpige 4

Stuepige 2

Udepige 2

